Pivotum

THE DEGREE RISK INDEX · FALL 2026 EDITION

So you're thinking about biology

The short version — written for you, not about you.

Everything that matters, none of the bits written for your parents.

We scored twenty-seven careers on how exposed they are to AI. Biology produced a result that surprises almost everyone.

Wet lab and bench research: 5.2 out of 10.

Bioinformatics and computational biology: 7.1.

The people at the bench are better protected than the people analysing what the bench produces. That's backwards from what most people assume, and it's the most useful thing here.

But there's a second finding that matters more for most students, and it's about money.

The 60-second version

- The computational, "tech-forward" end of biology is **more** exposed than hands-on lab work.
- Field and environmental science scores best (4.2). Regulated work — trials, quality, manufacturing — is protected too.
- **The bigger issue:** a biology bachelor's most commonly leads to technician work at around **$47,000**.
- The same field in industry averages **$140,000–$160,000**. That's a gap of about three times.
- Which means: this degree is a **gateway**, not a destination. Have a plan for step two.

The inversion, and why it happens

You'd assume the computational version of biology is the safe one. It's quantitative, well-paid, hard to get into, and obviously modern.

The framework says the opposite. Here's why.

Ask what each job actually consists of.

Bioinformatics takes structured data — sequences, expression matrices, imaging output — and produces analysis and code. Structured inputs. Computational outputs. No physical presence. No licensed accountability.

That's the exposed profile in *every* field we scored. And it doesn't stop being the exposed profile just because the subject matter is biology.

Wet lab work needs hands. Pipetting, cell culture, animal handling, sample prep, working out why an instrument is behaving strangely when nothing in the protocol covers it.

So computational biology is exposed for the same reason data analysis is exposed — because it is data analysis. The biology is the subject. The shape of the work is computation.

One important caveat. This is about the *analysis layer*, not about scientists. A computational biologist who designs the experiment, decides what question is worth asking and owns the interpretation is doing protected work. Someone running pipelines on other people's data to answer someone else's question isn't.

The distinction is judgment and ownership — same as everywhere else in this index.

The scores

TRACK	SCORE
Field ecology & environmental science	4.2
Clinical trials & regulatory affairs	4.6
Wet lab & bench research	5.2
Lab technician & diagnostics	5.8
Bioinformatics & computational biology	7.1
Scientific data curation & literature analysis	7.8

And the gap is *widening* — it was 1.2 points in 2023, it's 1.9 now. The computational end isn't converging with the bench. It's separating from it, in the direction nobody expected.

Now the part that matters more

A biology bachelor's on its own leads to technician work.

- Biological technicians: median around **$47,000–$48,000**
- Entry lab tech / research assistant: **$40,000–$55,000**
- Clinical research coordinator: median **$52,880**

- Environmental scientist: median near **$80,000**
- Forensic science technician: **$63,740**, with 11% growth

Now the same field in industry:

- National average base for life sciences professionals: **mid $140,000s to low $160,000s**
- Process engineers / manufacturing science: **$110,000–$160,000**
- Senior quality and compliance: **$90,000–$140,000**
- Commercial roles: **$110,000–$170,000** base

That's roughly three times, for the same underlying science.

The gap isn't about ability. It's about sector. Academic and bench-adjacent work is funded by grants. Industry is funded by products. The science overlaps a lot. The economics don't at all.

So the real question isn't "is biology safe"

It's: **what is this degree a gateway to?**

Most people who do well from a biology degree do something *next*:

- **Medicine, dentistry, veterinary, physician associate, pharmacy** — very common and very successful
- **Industry — regulatory affairs, quality assurance, clinical trial management, manufacturing.** Protected, well paid, and almost nobody applies to these from a biology degree.
- **A specialist master's** in something employable
- **Environmental and field science**, which scores best here and pays near $80,000

And the PhD question, honestly: it's worth a 40–50% salary uplift over a bachelor's, at the cost of four to seven years. That's a genuine return in industry. It's much weaker in academia, where far more people are trained for research careers than there are permanent positions.

Worth knowing that before you commit to the path, not during it.

What actually holds up

Doing the experiment well. The tacit craft that makes your results reproducible and someone else's not. Least automatable thing here and genuinely hard to teach.

Noticing something's wrong before any measurement tells you.

Regulated accountability. Clinical trials, GMP manufacturing and diagnostics all require a named responsible person who signs off. That's real protection *and* it's where the pay is — which is an unusually convenient overlap.

Experimental design — working out what would actually answer the question.

Interpreting the unexpected — the result that doesn't fit, which is where most real discovery happens.

One genuinely new thing

Hypothesis generation from existing data is now meaningfully assisted. That's the first properly intellectual part of science to be affected, and the profession hasn't settled what it means.

Designing the discriminating experiment and interpreting a surprising result are still human. Proposing candidate explanations increasingly isn't.

Which shifts where your value sits — toward asking the right question and judging whether an answer is believable, and away from producing the analysis.

Does this actually sound like you?

It suits people who:

- Are **genuinely curious about how living things work** — the subject has to be interesting on its own terms, because the early pay isn't compensating you
- Are **patient and precise** — most experiments fail
- Can **tolerate uncertainty** for long stretches
- Are **practically dextrous**, if you're aiming at bench work
- **Are willing to plan the next step**

Harder if you want strong early earnings, need certainty and closure, or assume the degree is the destination. That last one is the most common and most expensive mistake in this profile.

What actually matters when picking a course

- **How much hands-on lab or field time, and starting when?** This is the differentiator on applications and employers actually assess it.
- **Is statistics compulsory**, and taught as reasoning rather than as a service module?
- **Are there industry placements**, not just academic research ones? Given the pay gap, this is a big deal.
- **Does it cover regulated environments** — GMP, GLP, clinical trials? Rarely taught, highly employable.
- **Where are graduates at one year — by sector?**

Two things worth doing

Get real lab or field experience early. It's what separates applications, and it'll tell you quickly whether you enjoy the actual doing rather than the idea.

Then find someone three to five years past a biology degree and ask what they actually do and what they earn. The range will be wider than you expect, and understanding *why* is the whole lesson.

One last thing

Being fascinated by living things is a good reason to study this. It just isn't sufficient on its own.

The people who do well from a biology degree are almost always the ones who worked out early what it was a gateway to.

And if you're drawn to the computational side — keep your hands too. Being the person who can both run the analysis *and* design the experiment is a much stronger position than being able to do either alone.

If you want to go further — three things worth twenty minutes

- **Biological Technicians — Occupational Outlook Handbook** — the pay reality for the job most biology graduates enter first. Then read **Medical Scientists** and **Environmental Scientists** next to it.
- **BLS life and physical science occupations** — growth and pay for every track in this profile, comparable side by side.
- **US Life Sciences Salary Guide 2026** — industry pay by function. It's from a staffing firm, so read it for shape rather than gospel — but the gap against academic pay is the point.

There's a longer version behind this — full scores by track, how they were worked out, the financial picture in detail, what to ask a program, and where we might be wrong. Your parents have it.